MediSphere Cognitive Twin

AI Health Prediction Platform

8 Weeks | 4 Milestones

1. Project Idea

MediSphere creates digital health twins for patients and predicts future health risks using continuous learning models. The platform ingests data from FHIR(Fast Healthcare Interoperability Resources) APIs, wearables, and EHR systems via Kafka. MongoDB stores patient twins. TensorFlow Federated enables privacy-preserving risk prediction across hospitals. AI models predict cardiovascular risk, diabetes complications, and hospital readmission. The system enables preventive care through early intervention and personalized care plans while maintaining HIPAA compliance.

2. Emerging Skills

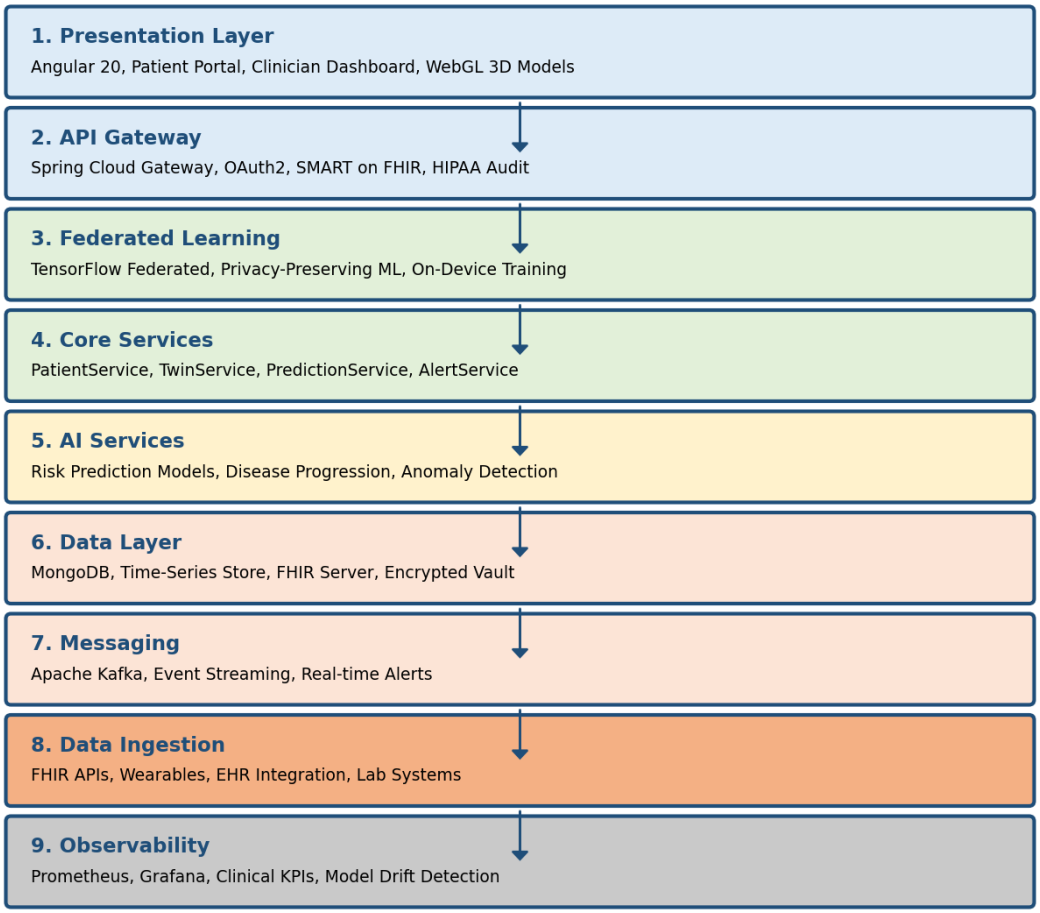
- Healthcare AI
- Digital Twin Technology
- Federated Learning
- FHIR Interoperability
- Clinical Decision Support
- HIPAA Compliance

3. Technology Stack

Component	Technology
Backend	Java 25, Spring Boot 4
Frontend	Angular 20, 3D Body Models
Database	MongoDB, Time-Series Store
Messaging	Apache Kafka
AI/ML	TensorFlow Federated
Healthcare	FHIR APIs, SMART on FHIR
Infra	Docker, Kubernetes, HIPAA Vault

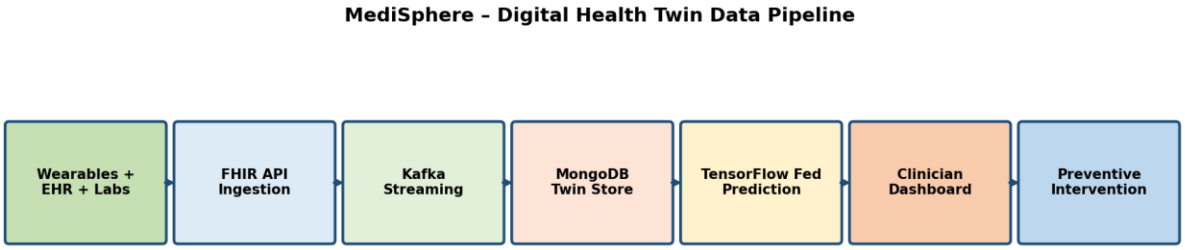
4. System Architecture Diagrams

MediSphere – 9-Layer Cognitive Health Twin Architecture



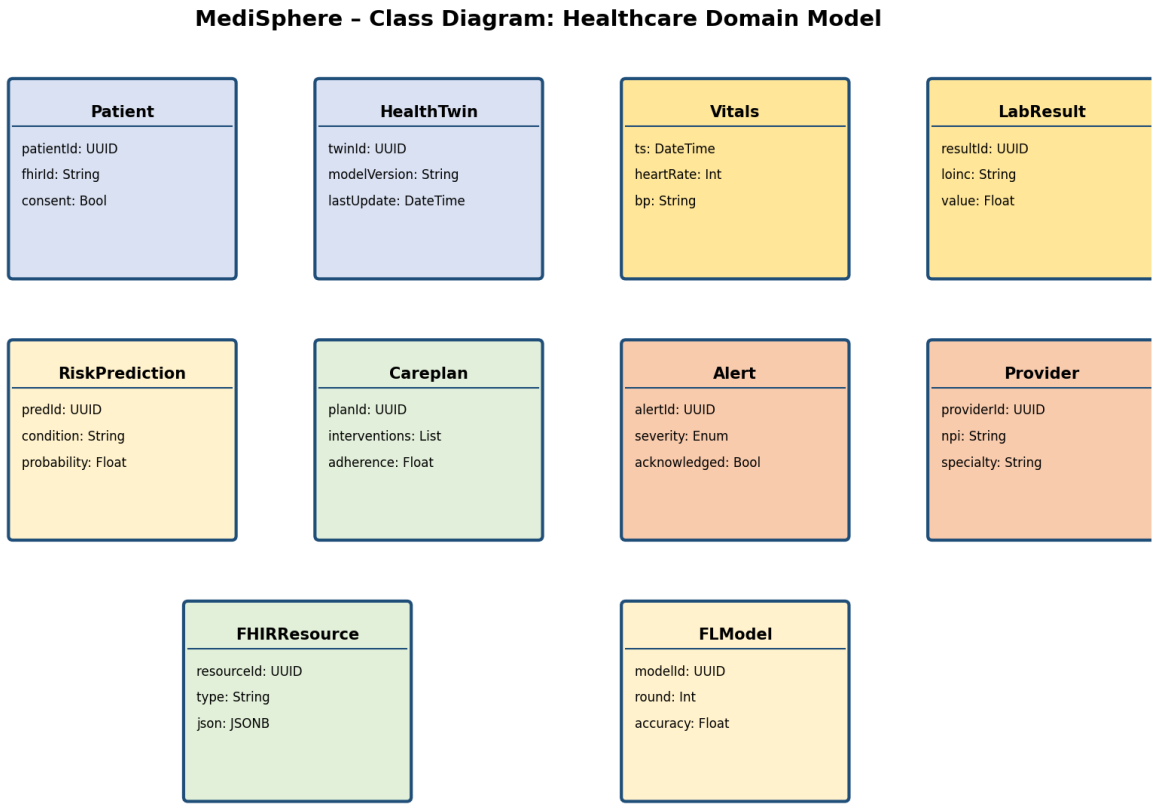
9-layer architecture with Angular patient portal, TensorFlow Federated for privacy-preserving ML, MongoDB for twins, and FHIR API ingestion.

5. Project Working Diagram



Data pipeline: Wearables + EHR + Labs → FHIR API → Kafka → MongoDB → TensorFlow Federated → Clinician Dashboard → Preventive Intervention.

6. Class Diagrams



Core entities: Patient, HealthTwin, Vitals, LabResult, RiskPrediction, Careplan, Alert, Provider, FHIRResource, FLModel.

7. Milestone-wise Deliverables & Screens

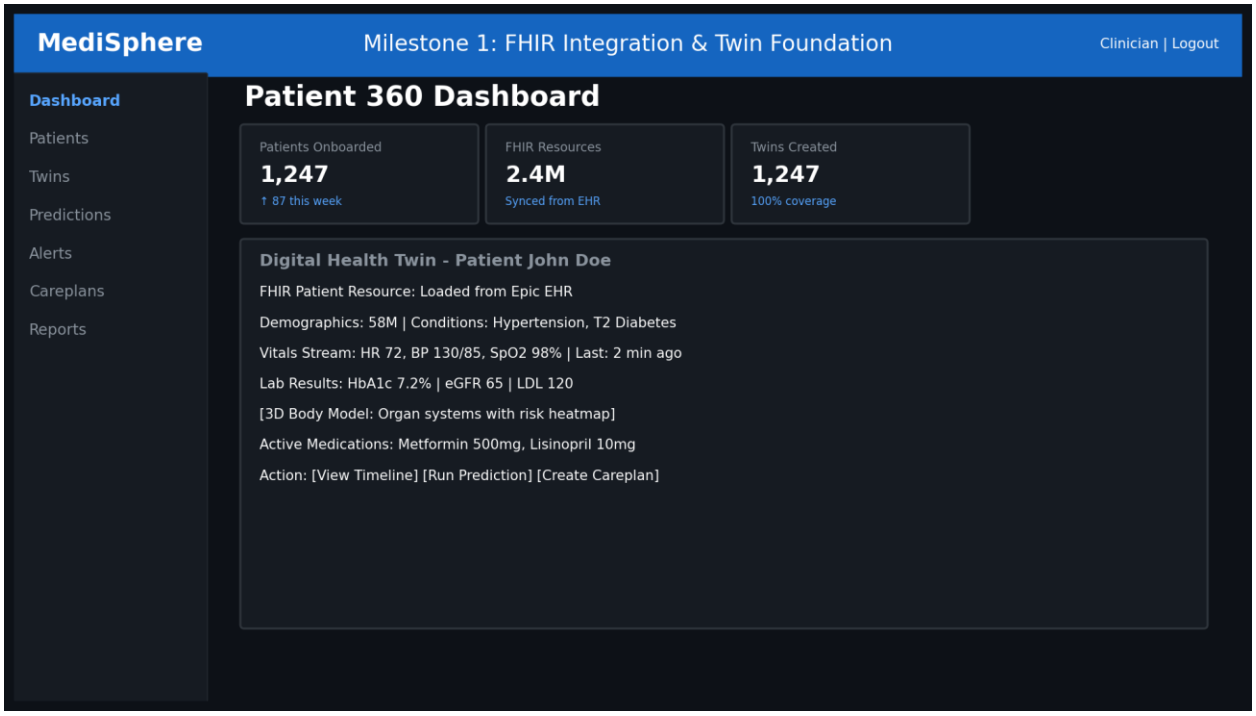
Milestone 1 (Weeks 1-2) – FHIR Integration & Twin Foundation

Implements FHIR API integration with EHR systems, creates digital health twins in MongoDB, and builds a patient 360 dashboard. Establishes Kafka streaming for real-time vitals from wearables. Implements HIPAA-compliant consent management.

Key Modules

- FHIR R4 API integration
- MongoDB patient twin store
- SMART on FHIR authentication
- Kafka vitals streaming
- Patient 360 UI
- Consent management

Expected Output Screen



Patient 360 Dashboard: 1,247 patients onboarded. 2.4M FHIR resources synced. Digital twin shows 3D body with risk heatmap.

Validation Screens

- FHIR resource validation
- HIPAA audit logging

- Patient consent verification
- Twin data completeness >95%
- Vitals range validation
- RBAC by provider/patient

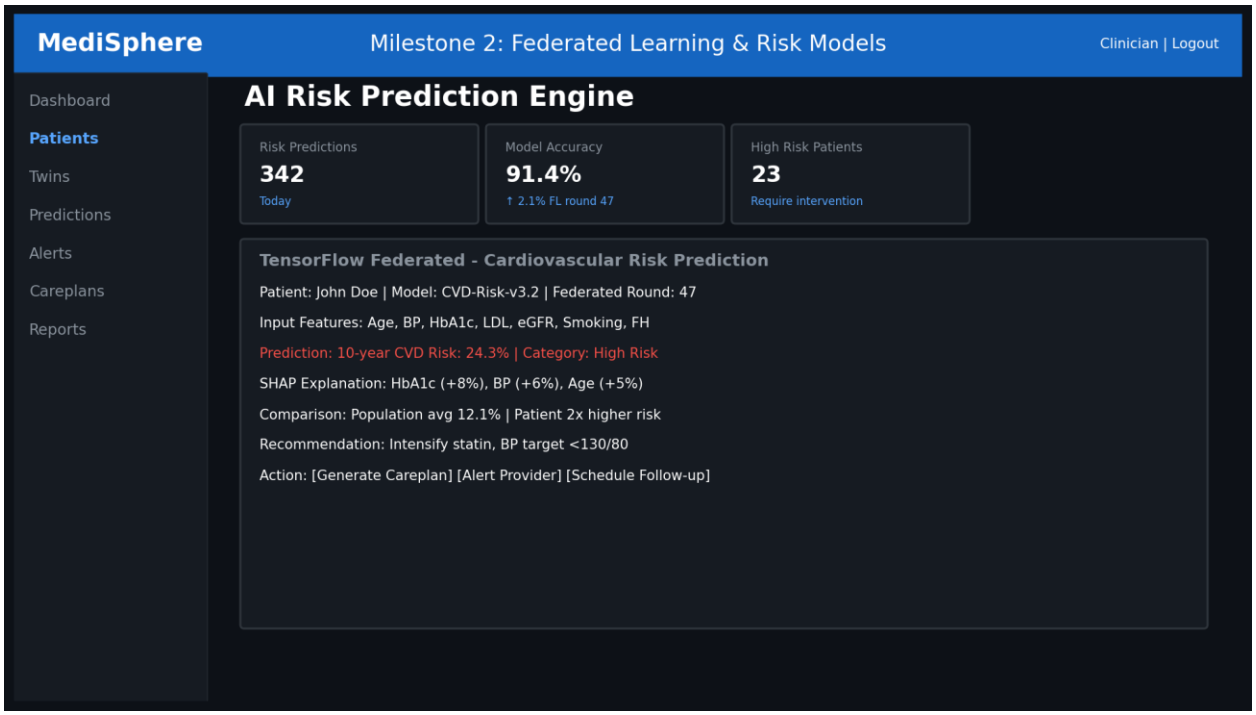
Milestone 2 (Weeks 3-4) – Federated Learning & Risk Models

Deploys TensorFlow Federated for privacy-preserving risk prediction. Models train across hospitals without sharing data. Achieves 91.4% accuracy for cardiovascular risk prediction with SHAP explainability.

Key Modules

- TensorFlow Federated setup
- CVD risk prediction model
- Diabetes complication model
- SHAP explainability
- Model versioning

Expected Output Screen



AI Risk Prediction: 10-year CVD risk 24.3% for John Doe. SHAP shows HbA1c +8%, BP +6%. Federated round 47 is complete.

Validation Screens

- Model accuracy >90%
- Federated round convergence

- SHAP explanation validity
- Prediction calibration
- Bias audit across demographics
- Clinical guideline compliance

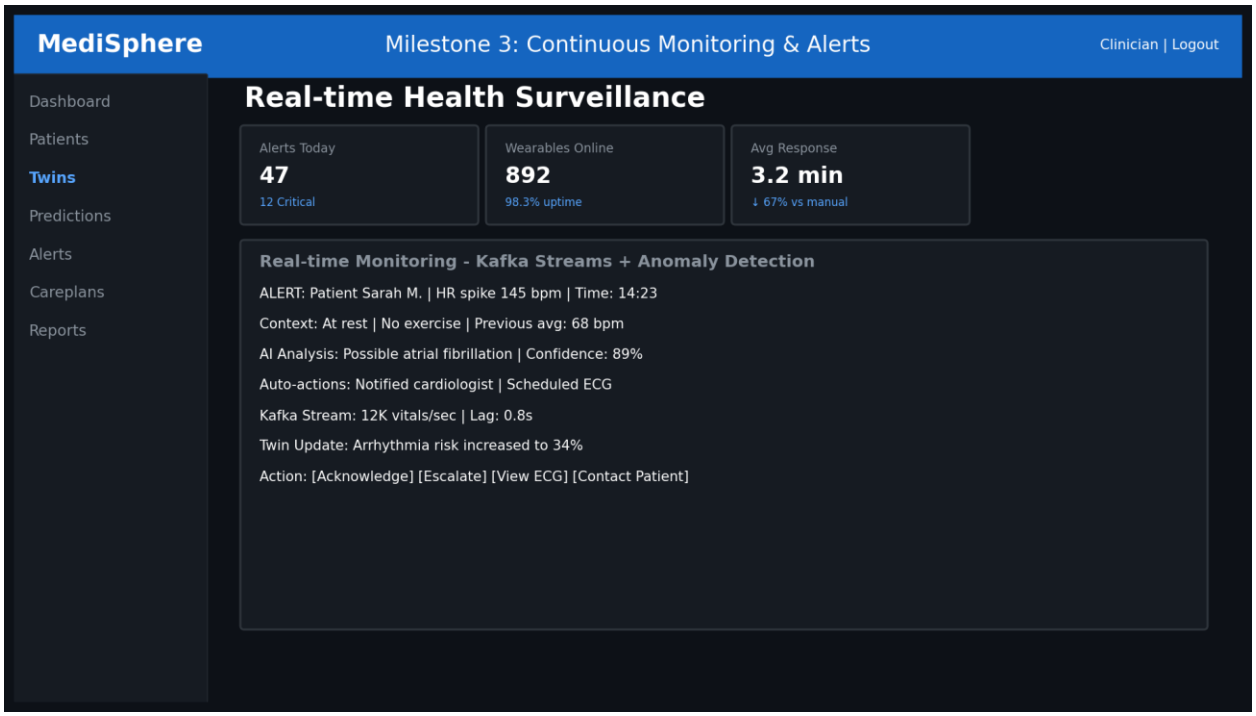
Milestone 3 (Weeks 5-6) – Continuous Monitoring & Alerts

Implements real-time monitoring from wearables via Kafka. AI anomaly detection triggers alerts. Reduces response time from hours to 3.2 minutes on average.

Key Modules

- Wearable device integration
- Kafka streams anomaly detection
- Real-time alert engine
- Clinical rule engine
- Mobile notifications

Expected Output Screen



Real-time Monitoring: Alert for Sarah M. - HR spike 145 bpm. AI analysis: Possible AFib with 89% confidence. Auto-notified cardiologist.

Validation Screens

- Vitals range validation
- Alert fatigue prevention
- Anomaly detection precision >85%

- Alert routing rules
- Acknowledgment tracking
- False alert rate <3%

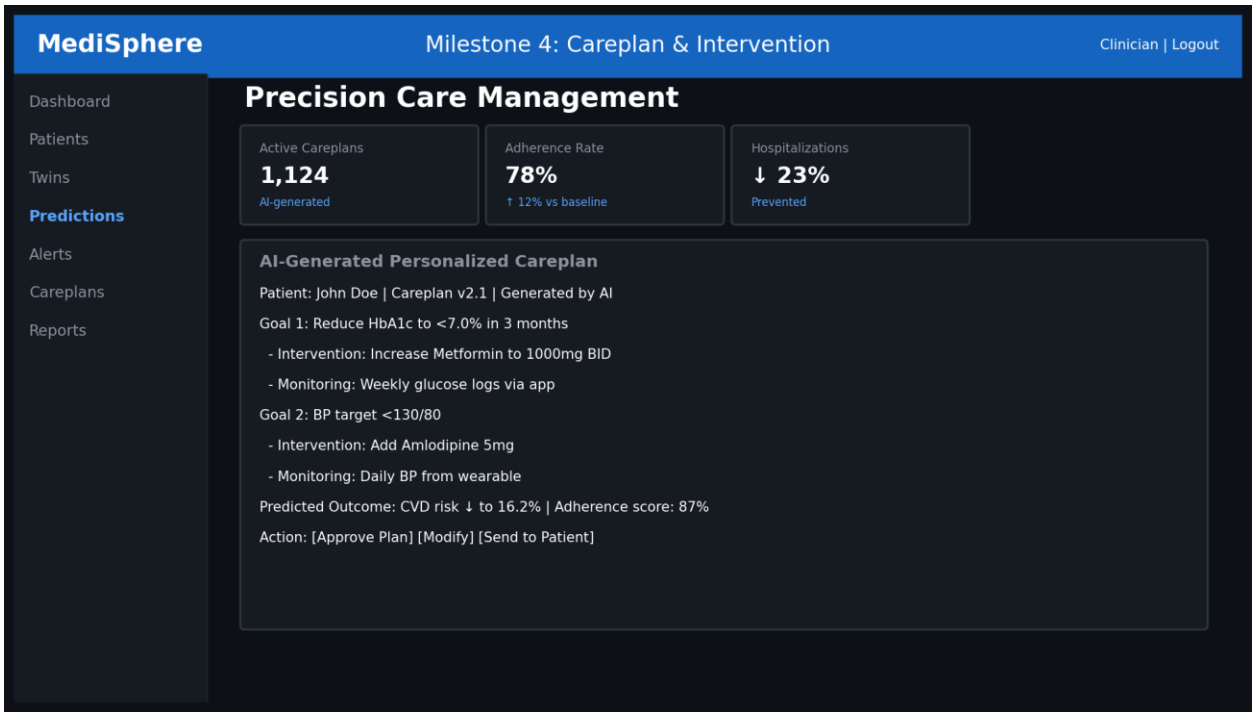
Milestone 4 (Weeks 7-8) – Careplan & Intervention

AI generates personalized care plans based on predictions and clinical guidelines. Tracks adherence and outcomes. Demonstrates 23% reduction in hospitalizations through preventive interventions.

Key Modules

- AI careplan generator
- Clinical guideline engine
- Adherence tracking
- Outcome measurement
- Provider collaboration

Expected Output Screen



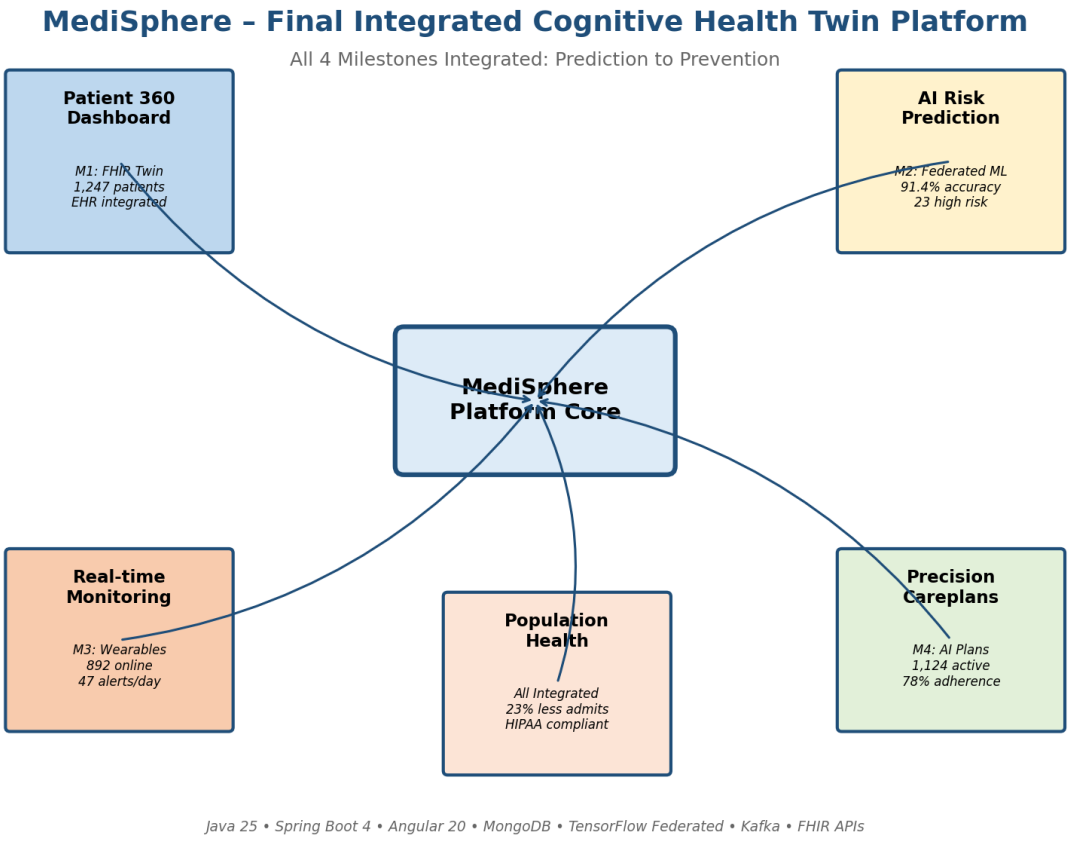
Precision Careplan: AI-generated plan for John Doe. Goal: HbA1c <7.0%. Interventions: Metformin increased, BP med added. Predicted CVD risk ↓ to 16.2%.

Validation Screens

- Clinical guideline compliance
- Careplan safety checks
- Drug interaction validation

- Adherence calculation accuracy
- Outcome tracking integrity
- Provider approval workflow

8. Master Application Screens – Final Integrated



Final integrated master screens after all 4 milestones:

- Patient 360 Dashboard: M1 – FHIR Twin, 1,247 patients, EHR integrated
- AI Risk Prediction: M2 – Federated ML, 91.4% accuracy, 23 high risk
- Real-time Monitoring: M3 – Wearables, 892 online, 47 alerts/day
- Precision Careplans: M4 – AI Plans, 1,124 active, 78% adherence
- Population Health: All integrated – 23% less admissions, HIPAA compliant

9. Validation Screens Framework

All screens enforce: FHIR resource validation, HIPAA audit logging, Patient consent verification, RBAC by provider/patient/role, comprehensive audit trail for all clinical decisions. AI predictions show confidence scores + SHAP explainability + clinical evidence.

Critical actions (medication changes, careplan approval) require provider signature. Federated learning ensures data never leaves hospital. All PHI encrypted at rest and in transit. Clinical decision support validated against guidelines. Bias audits across demographics.